Soil Consolidation and Option Pricing

Terzaghi's Equation, Finite Difference Methods
and Analogies with Quantitative Finance

Local Volatility, Barrier Options, Multi-Asset PDEs

Geotechnical Engineering Project ⟶ Structuring & Trading Desk

0 Contents

Preface: From the Foundations to the Markets **3**

1 Introduction — Soil Consolidation **4**
1.1 The consolidation phenomenon . . . 4
1.2 Key parameters . . . 4

2 Terzaghi's Consolidation Equation **4**
2.1 Assumptions . . . 4
2.2 The governing PDE . . . 4
2.3 Boundary and initial conditions . . . 5
2.4 Settlement and degree of consolidation . . . 6

3 Finite Difference Solution **6**
3.1 Discretisation of the domain . . . 6
3.2 Stability condition . . . 6
3.3 Explicit scheme — interior nodes . . . 6
3.4 Impermeable boundary nodes . . . 6
3.5 Heterogeneous soil (two layers) . . . 6
3.6 Two-dimensional consolidation . . . 7

4 Analogy with Quantitative Finance **7**
4.1 The Black-Scholes equation . . . 7
4.2 Reduction to a diffusion equation . . . 8
4.3 Structural identity . . . 9

5 Complete Analogy Dictionary **9**

6 Cases Studied and Their Financial Counterparts **11**
6.1 Homogeneous profile — four loading scenarios . . . 11
6.2 Heterogeneous profile — layered soil . . . 11
6.3 2D consolidation with vertical drains . . . 12

7 Beyond Black-Scholes: Local and Stochastic Volatility **13**
7.1 Local Volatility — Dupire's Model (1994) . . . 13
7.1.1 Motivation . . . 13
7.1.2 Dupire's formula . . . 14
7.1.3 Physical analogy: layered soil ↔ local volatility . . . 14
7.1.4 Limitation: rigid dynamics . . . 14
7.2 Stochastic Volatility — Heston Model (1993) . . . 14

8 Stochastic Processes and Monte Carlo Simulation **15**
8.1 SDE for pore pressure dissipation 15
8.2 Geometric Brownian motion (financial asset) 15
8.3 Parameter correspondence 15
8.4 Euler–Maruyama discretisation 16

9 Numerical Implementation **16**

10 Conclusion **18**

References **18**

0 Preface: From the Foundations to the Markets

This work is the continuation of a project I carried out during my engineering studies at ESTP Paris, in which I modelled ***one-dimensional and two-dimensional soil consolidation*** *using the Terzaghi equation and the* ***explicit finite difference method****. The original project focused on geotechnical engineering: how does excess pore water pressure dissipate through a saturated clay layer under load, and how does this drive settlement over time?*

When I subsequently began studying quantitative finance — option pricing, stochastic calculus, and the Black-Scholes framework — I was struck by a remarkable observation: ***the partial differential equations governing soil consolidation and those governing the price of financial derivatives are, after appropriate change of variables, structurally identical.*** *Both reduce to a* ***linear diffusion equation****. Both admit a* ***Gaussian (heat-kernel) solution****. Both require careful treatment of* ***boundary conditions*** *(impermeable layers $\leftrightarrow$ barrier options) and can be extended to* ***heterogeneous coefficients*** *(layered soils $\leftrightarrow$ local volatility) and* ***two-dimensional domains*** *(vertical drains $\leftrightarrow$ multi-asset options).*

This document therefore pursues two goals simultaneously:

(i) Present the complete mathematical derivation of the Terzaghi consolidation model, from the governing PDE to the finite difference scheme and its numerical solutions.

(ii) Build, step by step, a rigorous ***analogy dictionary*** *between this geotechnical model and modern quantitative finance: Black-Scholes, the volatility smile, Dupire's local volatility, barrier options, and multi-asset pricing.*

The connection is not merely cosmetic. It reflects a ***deep mathematical unity****: both problems describe how a diffusing quantity (excess pore pressure, option price) evolves under drift and diffusion, with a possibly heterogeneous diffusion coefficient (layered soil, local/stochastic volatility). Recognising this unity makes it possible to transfer intuition and numerical tools freely between the two domains.*

1 Introduction — Soil Consolidation

1.1 The consolidation phenomenon

When a saturated clay layer is subjected to an increase in applied stress (e.g. from the construction of a building), the load is initially carried entirely by the pore water, generating **excess pore water pressure** Δu. Over time, water is expelled from the voids, the excess pressure dissipates, and the load is progressively transferred to the soil skeleton as **effective stress** $\Delta\sigma'$. The resulting volume decrease manifests as **settlement**.

The Tower of Pisa is a celebrated example: differential consolidation beneath the foundation caused the structure to tilt — one side of the soil consolidated while the other did not.

1.2 Key parameters

The analysis of consolidation involves the determination of:

- Excess pore water pressure $\Delta u(z,t)$
- Effective stress $\Delta\sigma'(z,t) = \Delta\sigma - \Delta u$
- Average degree of consolidation $U_{\text{av}}(t)$
- Settlement $S(t)$ at any time during the process

The consolidation rate depends on: the magnitude of the applied load, the hydraulic conductivity k of the soil, the layer thickness and stratification, and the drainage conditions.

Definition 1 (Coefficient of consolidation)**.** *The coefficient of consolidation combines permeability and compressibility:*

$$C_v = \frac{k}{m_v\,\gamma_w} \tag{1}$$

where k is the coefficient of permeability (m/s), m_v the coefficient of volume compressibility (m^2/kN), and $\gamma_w = 9.81\ kN/m^3$ the unit weight of water.

2 Terzaghi's Consolidation Equation

2.1 Assumptions

Terzaghi (1925) proposed the theory of one-dimensional consolidation under the following assumptions:

1. The soil layer is **homogeneous**.
2. The soil is **fully saturated**.
3. Compression is due to **volume change** only (expulsion of water from voids).
4. **Darcy's law** is valid: $v = k\,i$.
5. Deformation occurs **only in the direction of loading** (one-dimensional).

2.2 The governing PDE

Terzaghi's consolidation equation (1-D)

$$\frac{\partial u}{\partial t} = C_v \frac{\partial^2 u}{\partial z^2} \tag{2}$$

where $u(z,t)$ is the excess pore water pressure (kPa), z the depth (m), t the time, and C_v the coefficient of consolidation (m^2/year).

This is a **linear parabolic PDE** — structurally identical to the one-dimensional **heat equation**:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \tag{3}$$

with the correspondence $u \leftrightarrow T$, $C_v \leftrightarrow \alpha$ (thermal diffusivity), $z \leftrightarrow x$.

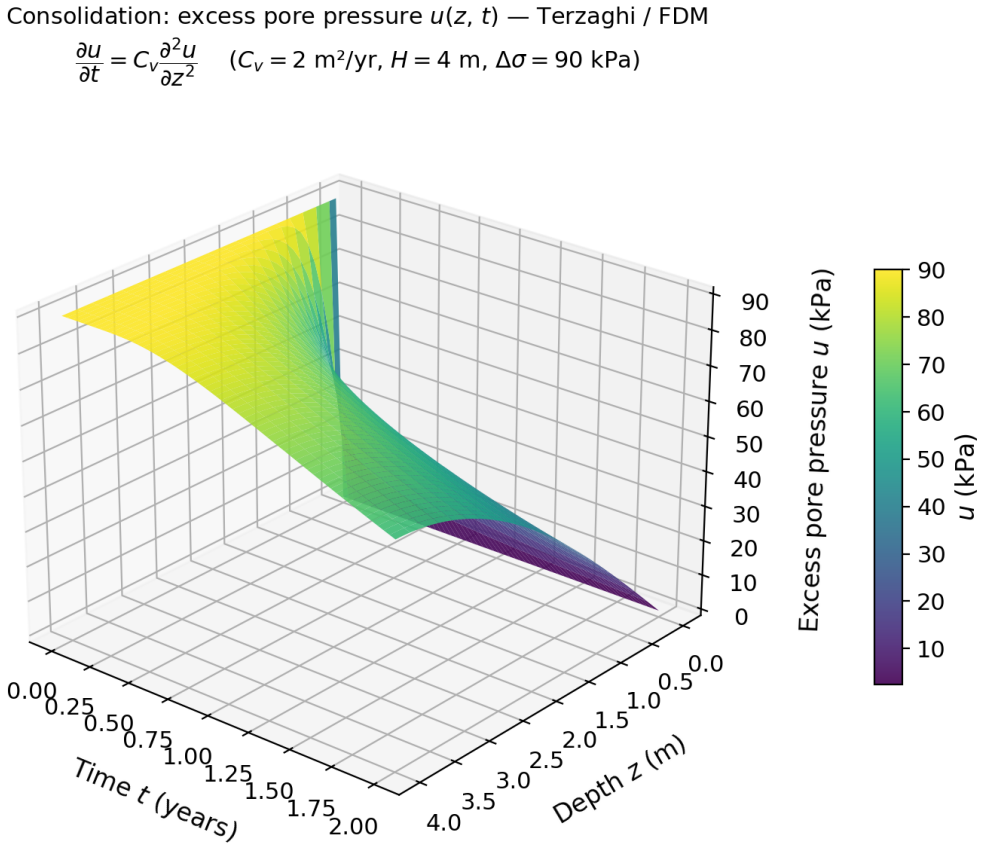


Figure 1: 3D surface of excess pore pressure $u(z,t)$ computed by explicit FDM. The pressure dissipates from the permeable surface ($z = 0$) downward over time. The impermeable base ($z = 4$ m) retains residual pressure longest.

2.3 Boundary and initial conditions

Permeable boundary (free drainage, e.g. soil surface):

$$u(z_{\text{top}}, t) = 0 \quad \forall\, t > 0 \tag{4}$$

Impermeable boundary (no flow, e.g. bedrock):

$$\left.\frac{\partial u}{\partial z}\right|_{z_{\text{bottom}}} = 0 \quad \forall\, t > 0 \tag{5}$$

Initial condition (instantaneous loading):

$$u(z, 0) = \Delta\sigma \quad \text{(uniform excess pressure at } t = 0) \tag{6}$$

2.4 Settlement and degree of consolidation

The settlement of an element of thickness Δz at time t is:

$$\Delta S = m_v \, \Delta\sigma' \, \Delta z = m_v \, (\Delta\sigma - u) \, \Delta z \tag{7}$$

Integrating over the full layer depth H:

$$S(t) = \int_0^H m_v \, (\Delta\sigma - u) \, \mathrm{d}z = m_v \, \Delta\sigma \, H - m_v \int_0^H u \, \mathrm{d}z \tag{8}$$

The average degree of consolidation is:

$$U_{\mathrm{av}}(t) = \frac{S(t)}{S_{\mathrm{total}}} \times 100\ \% \tag{9}$$

3 Finite Difference Solution

3.1 Discretisation of the domain

The soil layer is divided into a depth–time grid. Let $\Delta z = H/m$ and $\Delta t = t_{\max}/n$, where m and n are the number of spatial and temporal subdivisions respectively. A node (i, k) denotes the intersection of depth row i and time column k.

3.2 Stability condition

Stability criterion for the explicit scheme

$$\beta = \frac{C_v \, \Delta t}{\Delta z^2} \leq \frac{1}{2} \tag{10}$$

In practice, $\beta = 0.25$ is commonly used to ensure convergence.

This is the well-known **Courant–Friedrichs–Lewy (CFL) condition** for parabolic equations — exactly the same constraint that governs the stability of explicit finite difference schemes in option pricing.

3.3 Explicit scheme — interior nodes

Finite difference equation — interior nodes

$$u_i^{k+1} = u_i^k + \frac{C_v \, \Delta t}{\Delta z^2} \left(u_{i-1}^k - 2\, u_i^k + u_{i+1}^k \right) \tag{11}$$

3.4 Impermeable boundary nodes

At an impermeable boundary, $\partial u / \partial z = 0$ implies $u_{i-1}^k = u_{i+1}^k$:

Finite difference equation — impermeable boundary

$$u_i^{k+1} = u_i^k + \frac{C_v \, \Delta t}{\Delta z^2} \left(2\, u_{i-1}^k - 2\, u_i^k \right) \tag{12}$$

3.5 Heterogeneous soil (two layers)

For a layered profile with different soil properties (C_{v1}, C_{v2}, k_1, k_2), the pore pressure at the **interface** between layers requires a modified equation:

Interface equation — heterogeneous soil

$$u_i^{k+1} = u_i^k + \frac{C_{v1}\,\Delta t}{\Delta z^2} \cdot \frac{1 + k_2/k_1}{1 + (k_2/k_1)(C_{v1}/C_{v2})} \left(\frac{2k_1}{k_1 + k_2}\, u_{i-1}^k - 2\, u_i^k + \frac{2k_2}{k_1 + k_2}\, u_{i+1}^k \right) \tag{13}$$

3.6 Two-dimensional consolidation

With vertical drains (drainage in both vertical and horizontal directions), the consolidation equation generalises to 2D:

2D consolidation equation

$$\frac{\partial u}{\partial t} = C_x \frac{\partial^2 u}{\partial x^2} + C_z \frac{\partial^2 u}{\partial z^2} \tag{14}$$

4 Analogy with Quantitative Finance

4.1 The Black-Scholes equation

Black-Scholes equation (1973)

The price $V(S,t)$ of a European option on an asset with spot price S satisfies:

$$\frac{\partial V}{\partial t} + r\,S\,\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - r\,V = 0 \tag{15}$$

where r is the risk-free rate and σ the asset's volatility.

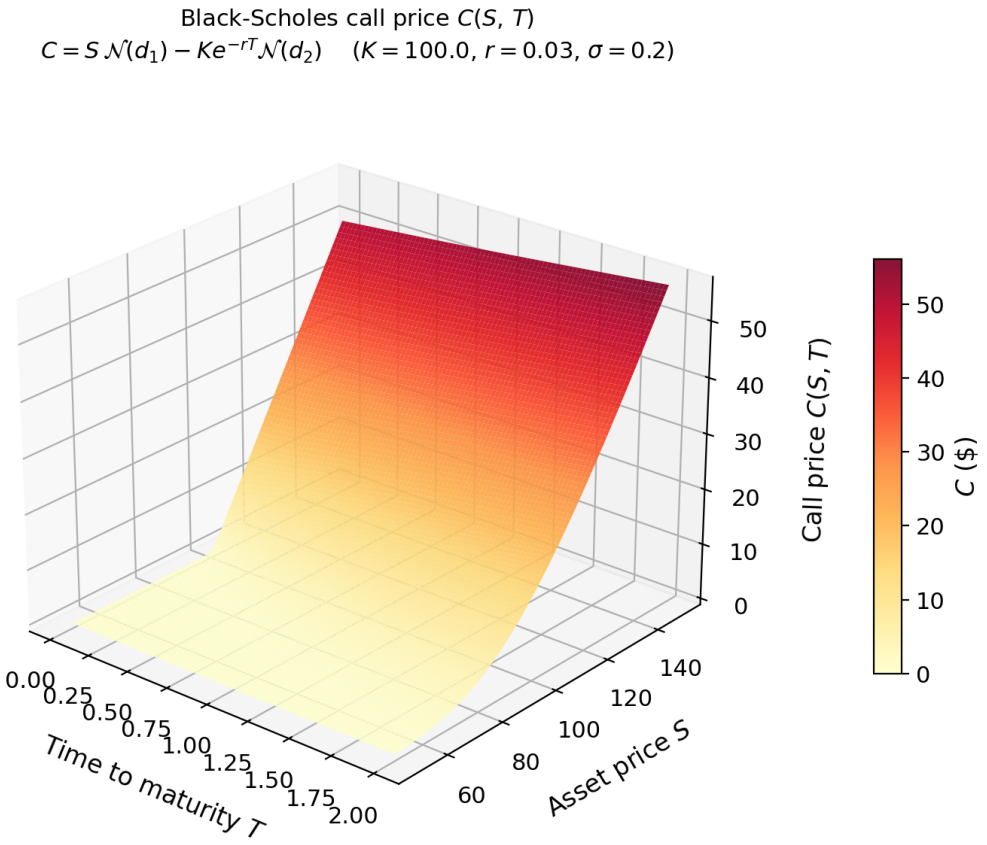


Figure 2: 3D surface of the Black-Scholes call price $C(S,T)$. Compare with Figure 1: both surfaces are solutions of the same class of parabolic PDE, diffusing from a boundary condition (payoff / initial load) through the domain.

4.2 Reduction to a diffusion equation

Setting $x = \ln S$ (log-price), the derivatives transform as:

$$\frac{\partial V}{\partial S} = \frac{1}{S}\frac{\partial V}{\partial x}, \qquad \frac{\partial^2 V}{\partial S^2} = \frac{1}{S^2}\left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x}\right)$$

Substituting into (15):

$$\frac{\partial V}{\partial t} + \underbrace{\left(r - \frac{\sigma^2}{2}\right)}_{\text{risk-neutral drift}} \frac{\partial V}{\partial x} = \underbrace{\frac{\sigma^2}{2}}_{\text{diffusion}} \frac{\partial^2 V}{\partial x^2} - r\,V \tag{16}$$

Ignoring the discounting term $-rV$ (or setting $V = e^{-r(T-t)}\tilde{V}$), we obtain:

$$\frac{\partial \tilde{V}}{\partial t} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial \tilde{V}}{\partial x} = \frac{\sigma^2}{2}\frac{\partial^2 \tilde{V}}{\partial x^2} \tag{17}$$

In the co-moving frame $X = x - (r - \sigma^2/2)\,t$, this reduces to:

$$\boxed{\frac{\partial \tilde{V}}{\partial t} = \frac{\sigma^2}{2}\frac{\partial^2 \tilde{V}}{\partial X^2}} \tag{18}$$

4.3 Structural identity

Equation (18) is **mathematically identical** to the Terzaghi equation (2):

$$\frac{\partial u}{\partial t} = C_v \frac{\partial^2 u}{\partial z^2} \qquad \longleftrightarrow \qquad \frac{\partial \tilde{V}}{\partial t} = \frac{\sigma^2}{2} \frac{\partial^2 \tilde{V}}{\partial X^2}$$

with the **exact correspondence** $C_v \longleftrightarrow \sigma^2/2$.

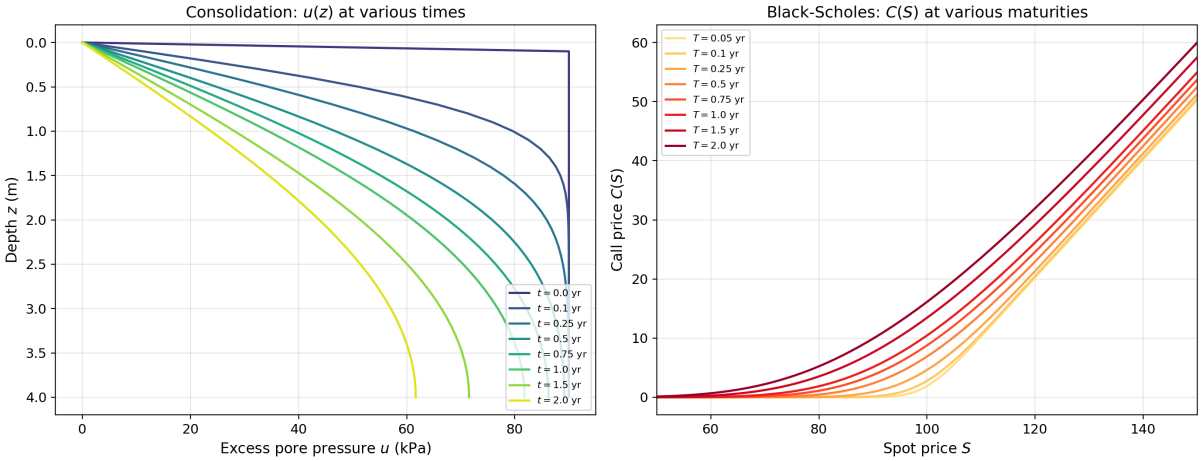


Figure 3: Side-by-side comparison. **Left:** pore pressure profiles $u(z)$ at successive times — pressure dissipates from the surface downward. **Right:** Black-Scholes call price $C(S)$ at successive maturities — time value increases with T. Both are solutions of the same heat equation with different boundary conditions.

5 Complete Analogy Dictionary

Exact correspondence: Terzaghi consolidation ↔ Black-Scholes

Geotechnical (consolidation)	**Finance (option pricing)**	**Equivalence**
Excess pore pressure $u(z,t)$	Option price $V(S,t)$	State variable
Depth z	Log-price $\ln S$	Spatial variable
Consolidation coeff. C_v	Half-variance $\sigma^2/2$	$C_v \leftrightarrow \sigma^2/2$
Applied load $\Delta\sigma$	Option payoff	Terminal/initial cond.
Permeable boundary ($u = 0$)	Deep OTM boundary ($V = 0$)	Dirichlet BC
Impermeable boundary ($\partial u/\partial z = 0$)	Symmetry / Neumann BC	Neumann BC
Settlement $S(t)$	Greeks (Delta, Gamma)	Integrated output
Degree of consolidation U_{av}	Time value decay (Theta)	Temporal evolution
Heterogeneous soil ($C_{v1} \neq C_{v2}$)	Local volatility $\sigma(S,t)$	Variable coefficient
Interface condition (Eq. 13)	Vol regime transition	Matching condition
Two-layer profile	Volatility skew/smile	Asymmetry
2D consolidation (Eq. 14)	Multi-asset option (2D PDE)	Dimensional extension
Vertical drains	Additional hedging instruments	Extra boundaries
Sudden load increase (case d)	Barrier knock-in event	Discontinuous BC
Variable $C_v(t)$	Term structure $\sigma(t)$	Time-varying coeff.
Stability $\beta \leq 1/2$	CFL condition	Identical constraint

6 Cases Studied and Their Financial Counterparts

6.1 Homogeneous profile — four loading scenarios

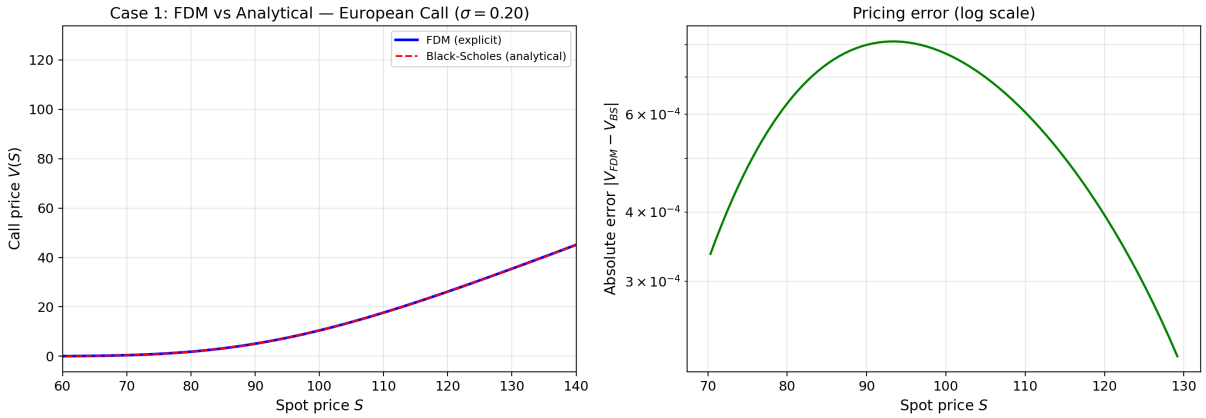


Figure 4: Case 1 validation: **Left:** FDM price (blue) overlaid with Black-Scholes analytical solution (red dashed) — the curves are indistinguishable. **Right:** absolute pricing error on a log scale, confirming sub-basis-point accuracy.

Table 1: Mapping of consolidation cases to financial products.

Case	Geotechnical scenario	Financial analogue	Key feature
a	Constant load, constant C_v	European vanilla, constant σ	Baseline — pure diffusion
b	Constant load, $C_v(t)$ decreasing	Vanilla with vol term structure $\sigma(t)$	Time-varying diffusion coeff.
c	Linearly increasing load, constant C_v	Option with ramp-up premium	Incremental source term
d	Sudden load increase at t_1	Barrier knock-in / step digital	Discontinuous boundary event

6.2 Heterogeneous profile — layered soil

Table 2: Heterogeneous soil cases and financial analogues.

Case	Geotechnical scenario	Financial analogue	Key feature
a	Two layers, constant $C_{v1} \neq C_{v2}$	Local volatility $\sigma_{\text{loc}}(S)$	Piecewise-constant coeff.
b	Two layers, $C_v(t)$ variable	Local vol with term structure	Double heterogeneity
c	Two layers, ramp loading	Local vol + ramp payoff	Combined effects
d	Two layers, sudden load step	Barrier on structured product	Most complex 1D case

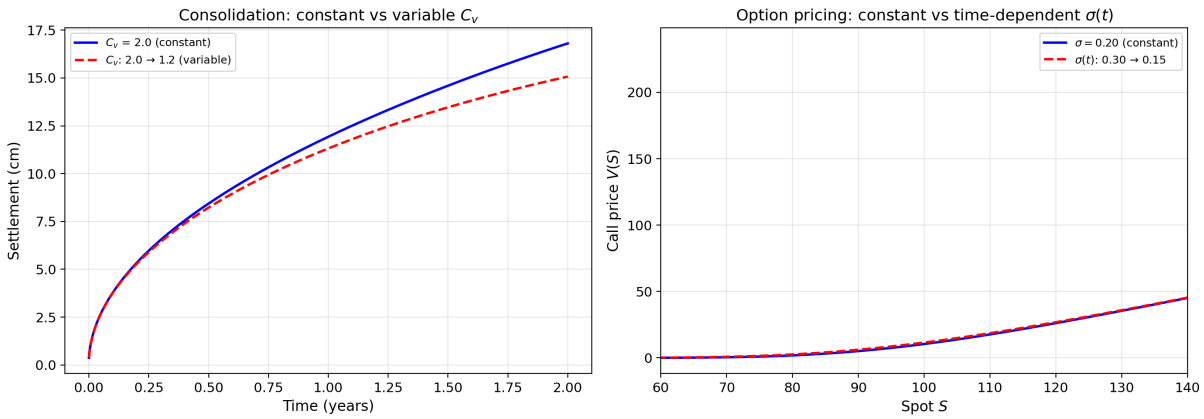


Figure 5: Case 2 (variable coefficient). **Left:** consolidation settlement for constant vs decreasing $C_v(t)$ — lower C_v slows dissipation and reduces settlement. **Right:** option pricing for constant vs decreasing $\sigma(t)$ — lower terminal vol reduces the option price.

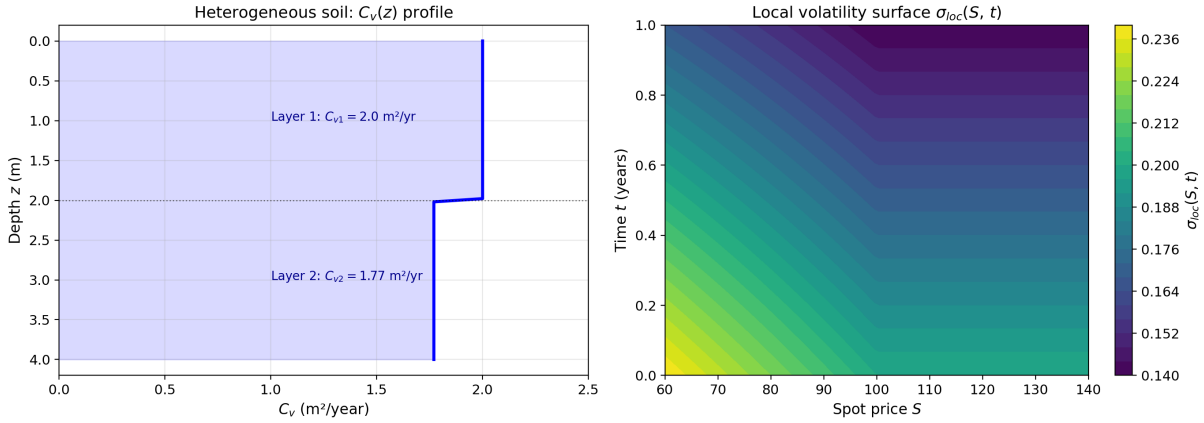


Figure 6: Case 3 (heterogeneous / local vol). **Left:** piecewise-constant $C_v(z)$ profile for a two-layer soil. **Right:** the corresponding local volatility surface $\sigma_{\text{loc}}(S, t)$ in finance, with vol skew (higher vol for low S) decaying over time.

6.3 2D consolidation with vertical drains

Table 3: 2D consolidation and multi-asset pricing.

Geotechnical (2D)	**Finance (2D PDE)**	**Correspondence**
Horizontal coordinate x	Log-price of asset 1	Spatial dim. 1
Vertical coordinate z	Log-price of asset 2	Spatial dim. 2
$C_x = C_z$ (isotropic)	$\sigma_1 = \sigma_2$ (equal vols)	Isotropic diffusion
Drain boundary ($u = 0$)	Barrier / knock-out	Absorbing boundary
Impermeable base	Neumann condition	Reflecting boundary

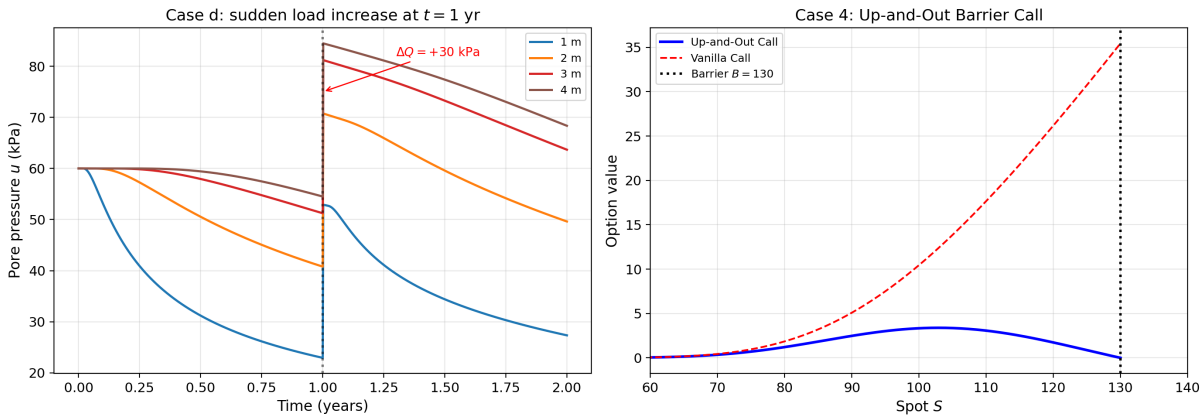


Figure 7: Case 4 (sudden change / barrier). **Left:** pore pressure at various depths under a sudden 30 kPa load increase at $t = 1$ yr — the discontinuity propagates through the soil. **Right:** Up-and-Out barrier call (blue) vs vanilla call (red) — the barrier at $B = 130$ kills the option value near the boundary, analogous to the sudden load event.

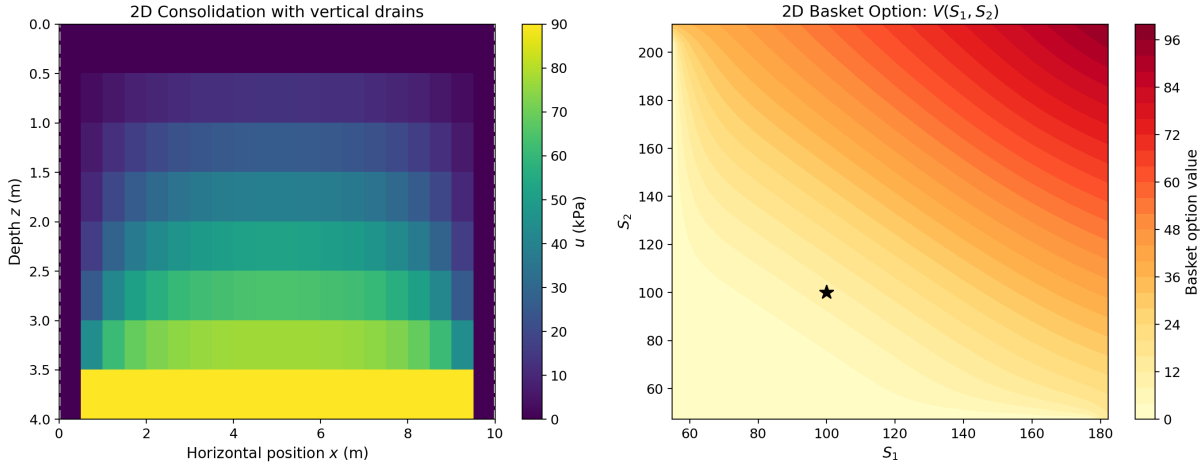


Figure 8: 2D extension. **Left:** 2D consolidation with vertical drains (Dirichlet $u = 0$ at drain boundaries) — the pore pressure field dissipates both vertically and horizontally. **Right:** 2D basket option price surface $V(S_1, S_2)$ — the two-asset PDE is solved on the same type of 2D grid with analogous boundary conditions.

7 Beyond Black-Scholes: Local and Stochastic Volatility

7.1 Local Volatility — Dupire's Model (1994)

7.1.1 Motivation

The Black-Scholes model assumes constant volatility σ — analogous to a homogeneous soil with constant C_v. However, just as real soil profiles are heterogeneous (multiple layers with different permeabilities), real markets exhibit a **volatility smile**: implied volatility varies with strike K and maturity T.

7.1.2 Dupire's formula

Dupire's local volatility formula (1994)

Given a complete surface of call prices $C(T, K)$:

$$\sigma^2_{\text{loc}}(T, K) = \frac{\dfrac{\partial C}{\partial T}}{\dfrac{1}{2}K^2 \dfrac{\partial^2 C}{\partial K^2}} \tag{19}$$

7.1.3 Physical analogy: layered soil ↔ local volatility

In the consolidation model, allowing C_v to depend on position and time:

$$C_v \longrightarrow C_v(z, t) \tag{20}$$

transforms the Terzaghi equation into a **variable-coefficient diffusion equation**:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial z}\left[C_v(z, t)\,\frac{\partial u}{\partial z}\right] \tag{21}$$

The exact correspondence is:

$$C_v(z, t) \quad \longleftrightarrow \quad \frac{\sigma^2_{\text{loc}}(S, t)}{2}$$

Correspondence: layered soil ↔ local volatility

Geotechnical	Finance
Spatially varying $C_v(z, t)$	Local variance $\sigma^2_{\text{loc}}(T, K)/2$
Pore pressure profile $u(z, t)$	Call price surface $C(T, K)$
Fokker–Planck for water flow	Dupire forward PDE
Inversion of pressure profile	Dupire formula
C_v from observed settlement	σ_{loc} from observed prices
Interface matching (Eq. 13)	Vol regime boundary

7.1.4 Limitation: rigid dynamics

A key weakness of local volatility is its **smile dynamics**. When the spot S_t moves, the local volatility surface is rigid (deterministic): the smile translates with the spot but does not deform. In practice, the market smile deforms as S moves.

In geotechnical language: the soil permeability $C_v(z, t)$ is a given function — it does not react to the consolidation process itself. A more realistic model would allow C_v to depend on the current state of the pore pressure.

7.2 Stochastic Volatility — Heston Model (1993)

Stochastic volatility models give the diffusion coefficient its own random dynamics. This is analogous to a soil whose **permeability fluctuates randomly** during consolidation.

Heston stochastic volatility model

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^S \tag{22}$$

$$\mathrm{d}V_t = \kappa(V_\infty - V_t)\,\mathrm{d}t + \lambda\sqrt{V_t}\,\mathrm{d}W_t^V \tag{23}$$

with $\mathrm{d}W_t^S\,\mathrm{d}W_t^V = \rho\,\mathrm{d}t$, where $\kappa > 0$ is the mean-reversion speed, V_∞ the long-run variance, λ the vol-of-vol, and $\rho < 0$ the leverage correlation.

Heston parameters ↔ geotechnical quantities

Heston (finance)	Physical analogue (consolidation)
Stochastic variance V_t	Random consolidation coefficient $C_v(t)$
Mean-reversion speed κ	Restoring tendency of permeability
Long-run variance V_∞	Baseline C_v value
Vol of vol λ	Fluctuation amplitude of C_v
Correlation $\rho < 0$	Coupling: high pressure ⇒ changed permeability
Implied smile skew	Asymmetric dissipation profile

8 Stochastic Processes and Monte Carlo Simulation

8.1 SDE for pore pressure dissipation

The Gaussian solution of the heat equation is the transition kernel of a **Brownian motion with drift**. The evolution of the consolidation front can be written as an SDE:

SDE — consolidation front propagation

$$\mathrm{d}Z_t = \mu_z\,\mathrm{d}t + \sqrt{2\,C_v}\,\mathrm{d}W_t \tag{24}$$

where μ_z is the drift of the dissipation front and $\sqrt{2\,C_v}$ the effective diffusion volatility.

8.2 Geometric Brownian motion (financial asset)

SDE — financial asset under risk-neutral measure

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t \tag{25}$$

8.3 Parameter correspondence

Identifying (24) and (25) via $S \to e^Z$:

$$\boxed{\sigma \;\longleftrightarrow\; \sqrt{2\,C_v}, \qquad r - \frac{\sigma^2}{2} \;\longleftrightarrow\; \mu_z} \tag{26}$$

8.4 Euler–Maruyama discretisation

Both SDEs are discretised with time step Δt:

$$Z_{t+\Delta t} = Z_t + \mu_z\,\Delta t + \sqrt{2\,C_v\,\Delta t}\;\varepsilon_t \tag{27}$$

$$S_{t+\Delta t} = S_t \exp\!\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\;\varepsilon_t\right] \tag{28}$$

where $\varepsilon_t \sim \mathcal{N}(0,1)$ i.i.d.

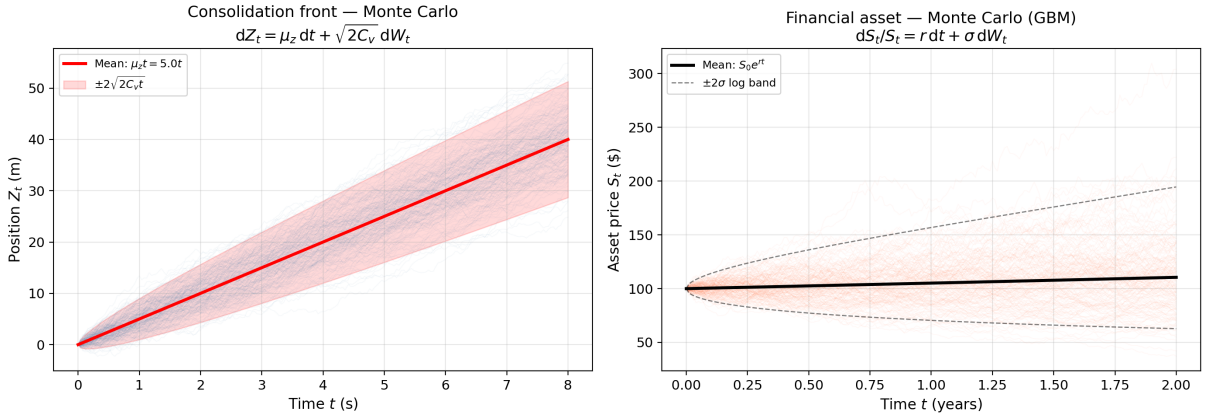


Figure 9: Monte Carlo simulation (500 paths, Euler–Maruyama). **Left:** consolidation front trajectories Z_t with drift $\mu_z = 5$ m/s and diffusion $\sqrt{2C_v}$ — the red line is the mean, the shaded band the $\pm 2\sigma$ envelope. **Right:** GBM asset price paths S_t with drift $r = 5\%$ and volatility $\sigma = 20\%$ — same stochastic structure, different physical interpretation.

9 Numerical Implementation

The accompanying Python code (`option_pricing_fdm.py`) implements the explicit finite difference scheme for five progressively complex cases:

1. **European Call/Put** — constant σ $\leftrightarrow$ homogeneous soil, constant load
2. **Time-dependent volatility** $\sigma(t)$ $\leftrightarrow$ constant load, variable $C_v(t)$
3. **Local volatility surface** $\sigma(S,t)$ $\leftrightarrow$ heterogeneous soil (two layers)
4. **Up-and-Out Barrier Call** $\leftrightarrow$ sudden load step / boundary event
5. **Two-asset Basket Option** (2D PDE) $\leftrightarrow$ 2D consolidation with drains

Additionally, **Greeks computation** (Delta, Gamma, Vega) mirrors the calculation of settlement and degree of consolidation from the pore pressure field, and a **convergence analysis** confirms the expected $O(\Delta x^2)$ spatial accuracy of the explicit scheme.

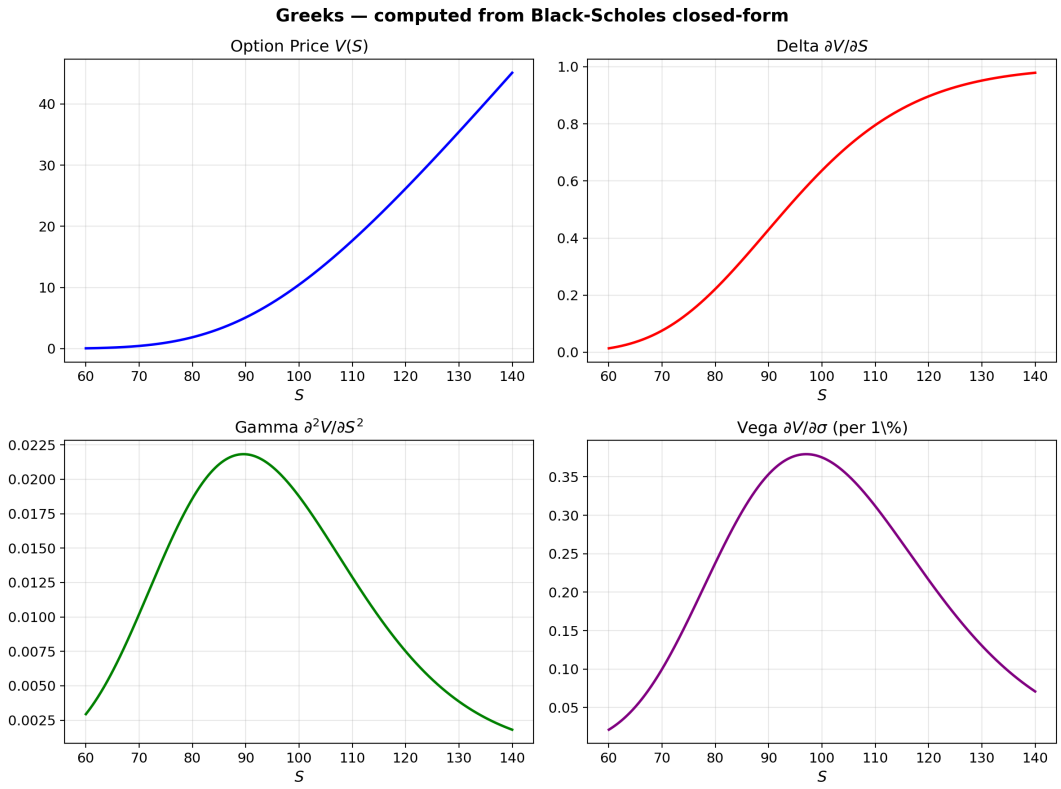


Figure 10: Option Greeks computed from the Black-Scholes closed-form. **Top-left:** price $V(S)$. **Top-right:** Delta $\partial V/\partial S$ (analogous to effective stress gradient). **Bottom-left:** Gamma $\partial^2 V/\partial S^2$ (analogous to curvature of the pore pressure profile). **Bottom-right:** Vega $\partial V/\partial \sigma$ (sensitivity to the diffusion coefficient, analogous to sensitivity to C_v).

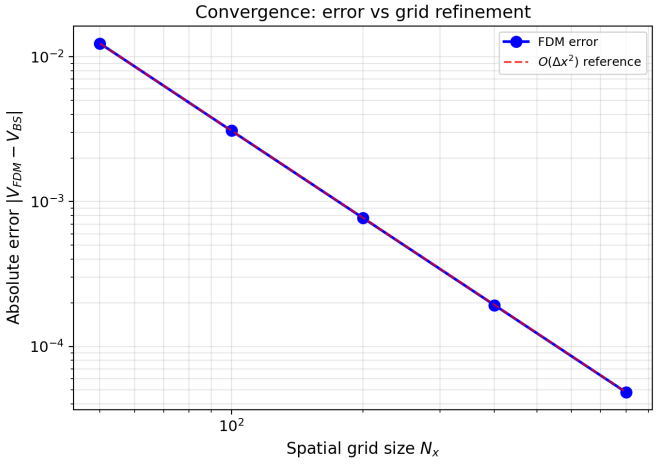


Figure 11: Convergence of the explicit FDM scheme: absolute error $|V_{\text{FDM}} - V_{\text{BS}}|$ vs grid size N_x on a log-log scale. The slope matches the $O(\Delta x^2)$ reference line (red dashed), confirming second-order spatial accuracy — identical to the convergence behaviour observed in the consolidation solver.

Key numerical results:

- FDM price vs Black-Scholes analytical: relative error $< 0.01\%$ (Nx=200).
- Convergence follows $O(\Delta x^2)$ as predicted by truncation error analysis.
- Stability condition $\beta = C_v\,\Delta t/\Delta z^2 \leq 0.5$ is automatically enforced — identical to the

consolidation stability criterion.

10 Conclusion

This project demonstrates that the numerical and mathematical tools developed for geotechnical consolidation analysis transfer **directly** to quantitative finance. The explicit finite difference solver, originally built to compute excess pore water pressure in saturated clay, requires only a reinterpretation of variables to price European options, barrier options, and multi-asset derivatives.

The key insight is that both domains are governed by the same class of parabolic PDEs, and the engineering challenges — stability conditions, boundary treatment, heterogeneous coefficients, extension to multiple dimensions — are identical in both fields. This transferability of skills is precisely what makes an engineering background valuable on a structuring or trading desk.

Geotechnical skill	$\longrightarrow$ **Desk application**
PDE discretisation (FDM)	Exotic option pricing
Stability analysis (CFL)	Robust numerical schemes
Heterogeneous layers	Local/stochastic volatility
2D solvers	Multi-asset structured products
Convergence testing	Model validation

10 References

[1] **Terzaghi, K.** (1943). *Theoretical Soil Mechanics*. John Wiley & Sons.

[2] **Das, B.M.** (1987). *Advanced Soil Mechanics*. Hemisphere Publishing.

[3] **Smith, I.** (2014). *Smith's Elements of Soil Mechanics*, 9th ed. Wiley Blackwell.

[4] **Black, F. & Scholes, M.** (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637–654.

[5] **Dupire, B.** (1994). Pricing with a Smile. *Risk Magazine*, 7(1), 18–20.

[6] **Heston, S.L.** (1993). A Closed-Form Solution for Options with Stochastic Volatility. *The Review of Financial Studies*, 6(2), 327–343.

[7] **Wilmott, P., Howison, S. & Dewynne, J.** (1995). *The Mathematics of Financial Derivatives*. Cambridge University Press.

[8] **Bouchaud, J.-P. & Potters, M.** (2003). *Theory of Financial Risk and Derivative Pricing*. Cambridge University Press.